Bayesian Hierarchical Modeling Of Log Returns And Portfolio Risk Analysis

Team Member(s):
Abhiram Vadlamani (avadlam5@ncsu.edu)

Special Thanks to:
Dr Sujit Ghosh

Outline

Introduction

Exploratory Data Analysis

The Hierarchical MCMC model

Copula Analysis

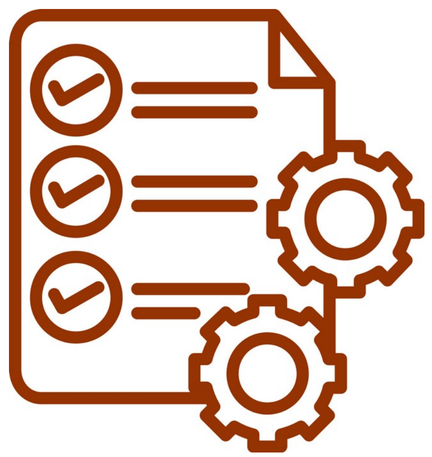
Results and Conclusion

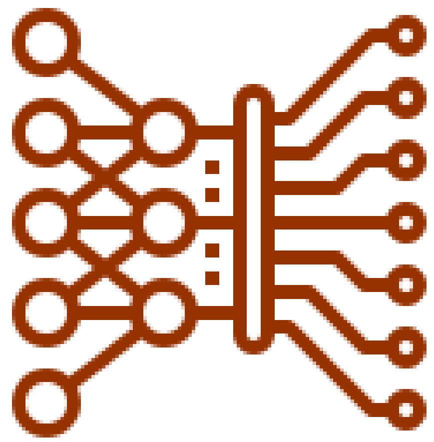
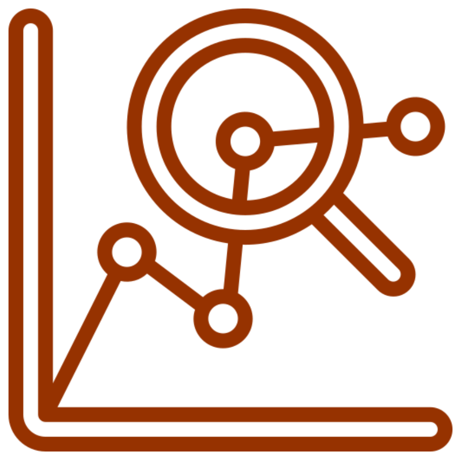
Motivation

- Quantifying risks and uncertainty in financial markets.
- A well fitted distribution allows for forecasting and scenario analysis.
- A well fitted distribution aids in the creation of trading strategies.
- Exhibiting why the Skewed t is a better fit for log returns than the normal.

Process Outline

Data Collection → Exploratory Data Analysis → Model Fitting and Copula Analysis → Model Assessment

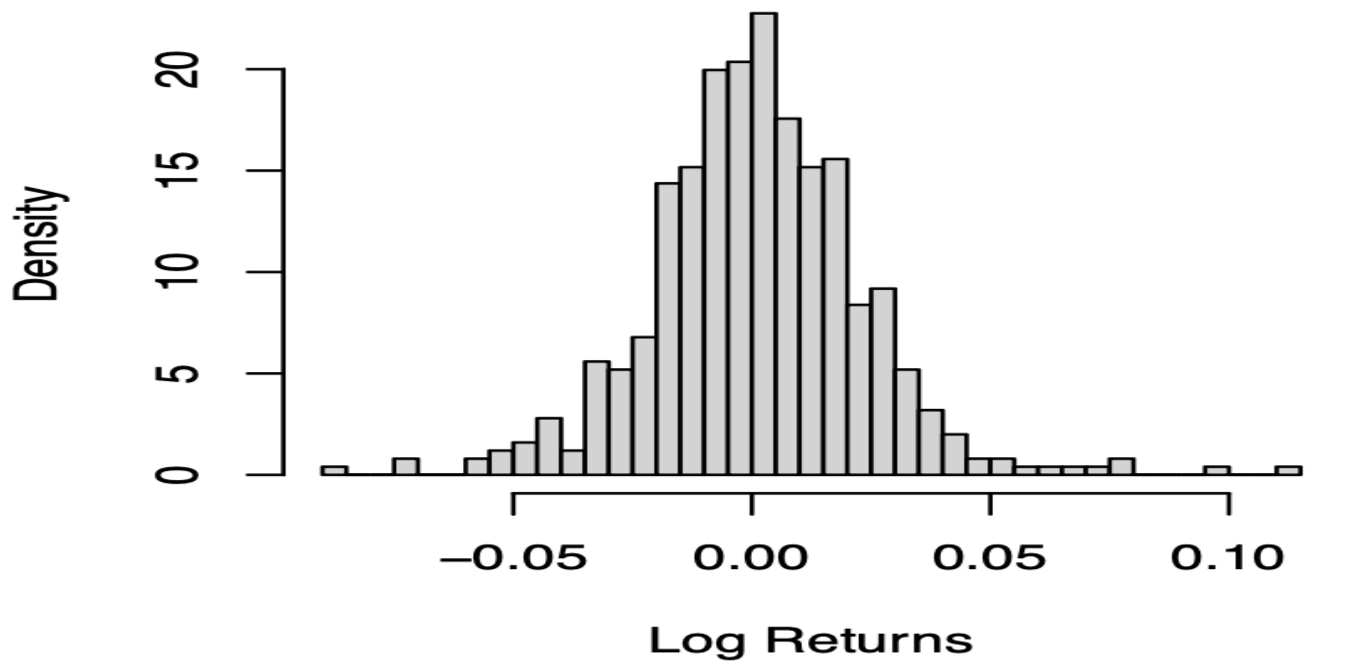


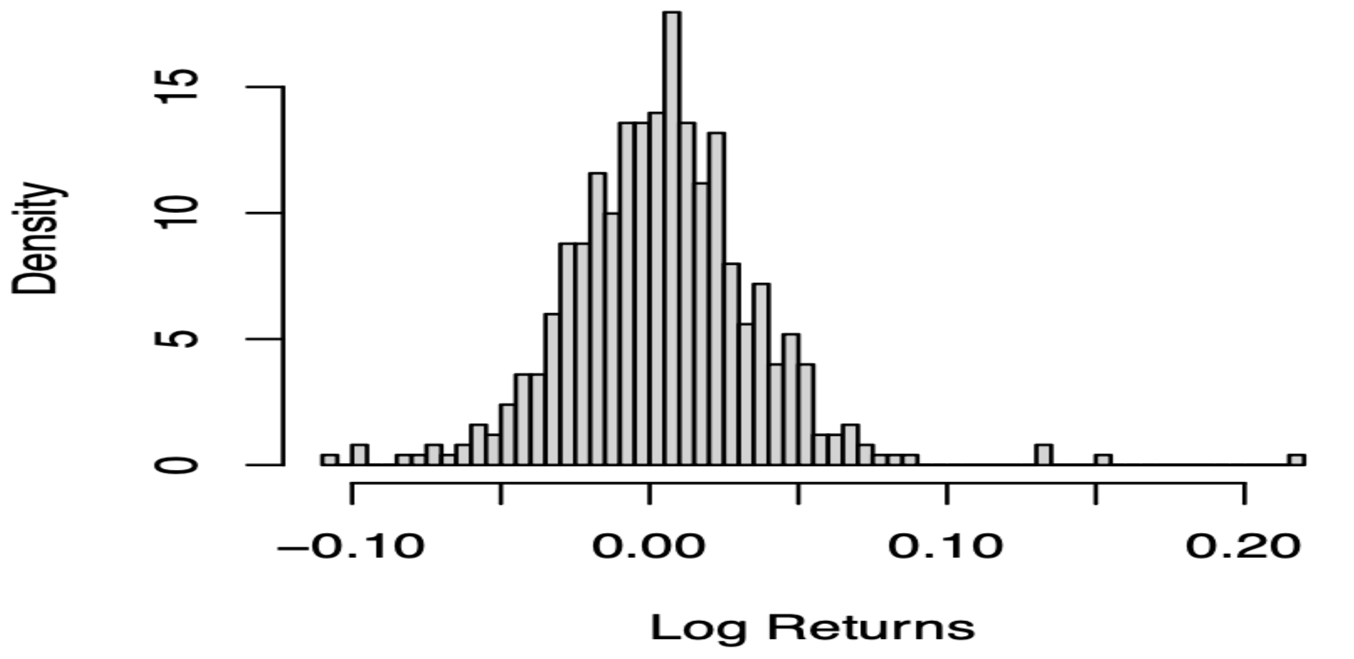
Exploratory Data Analysis (EDA)

Analyzed Stocks → Fit Distributions → Goodness of fit tests → Assessment

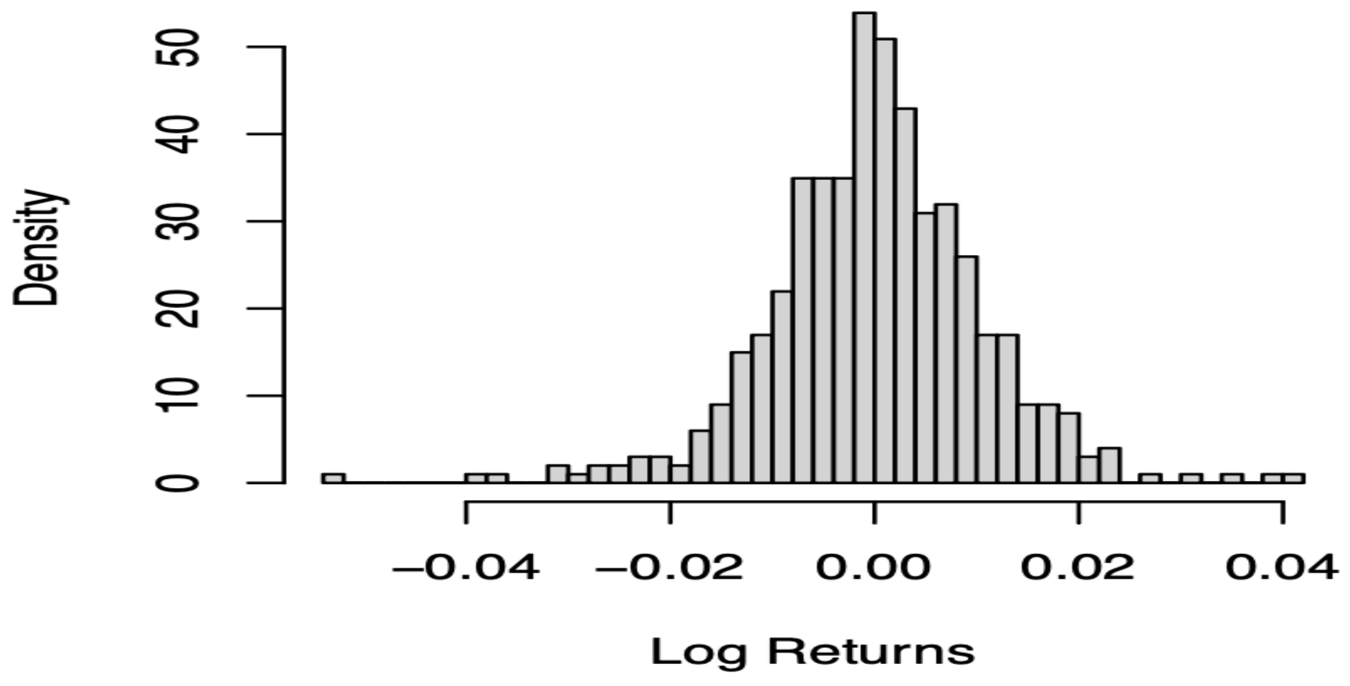
Log Returns Distribution



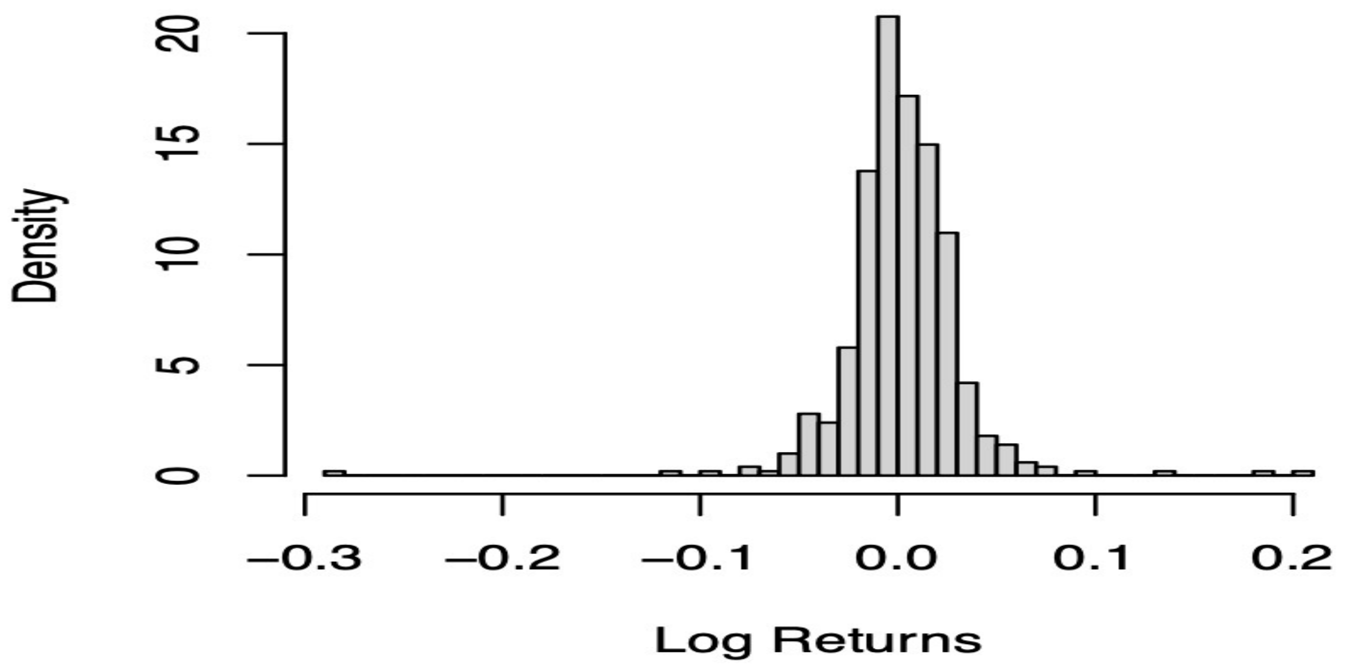
AMZN



NVDA



PEP



META

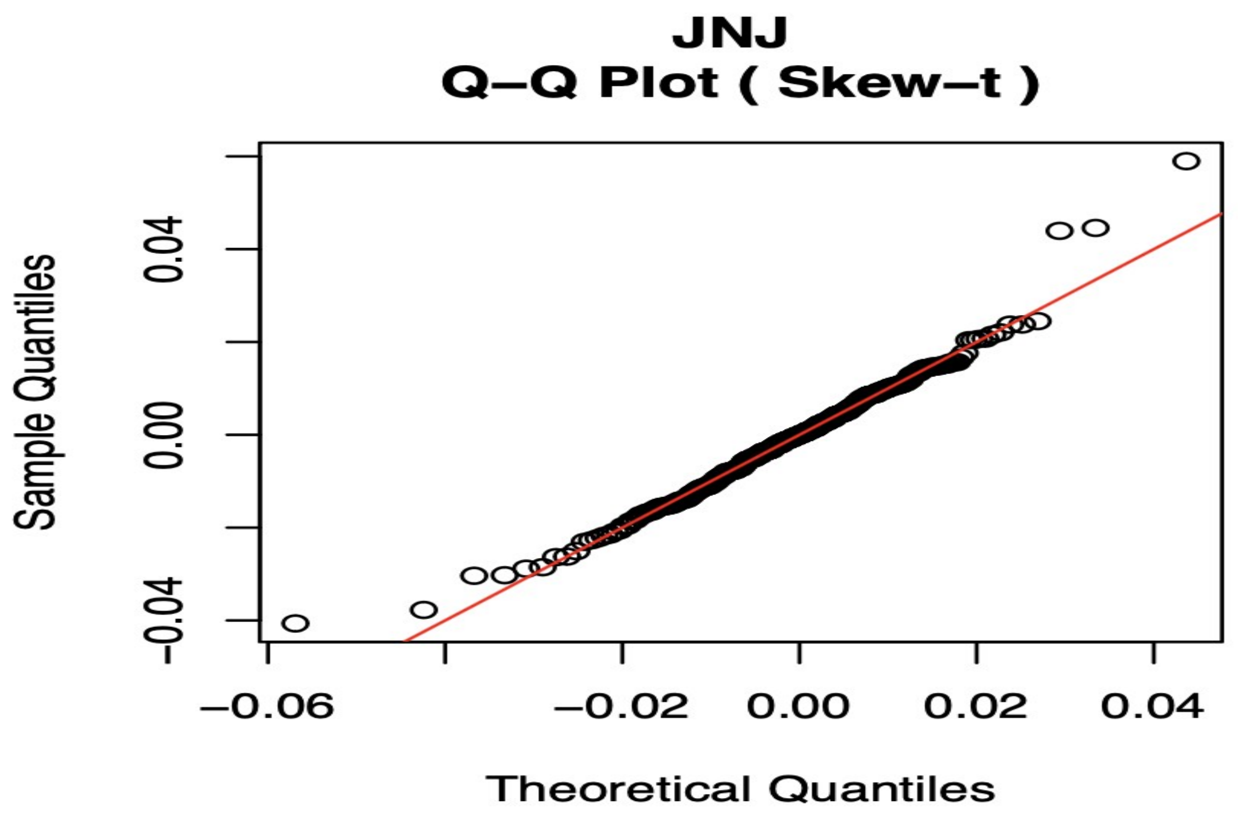
EDA (Continued) – Normal

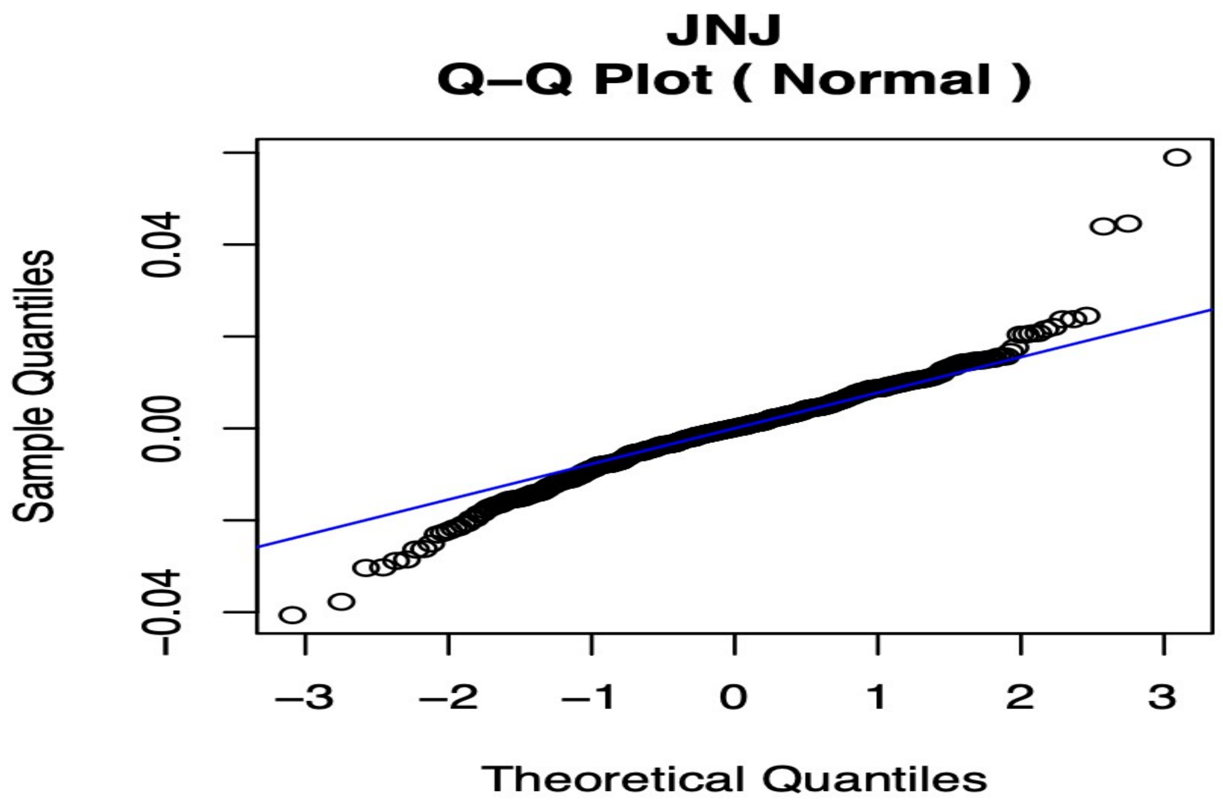
Stock (Ticker)	Kolmogorov – Smirnov Test p – value	Anderson Darling Test p – value
MSFT	0.182	0.09
JNJ	0.029	0.014
PG	0.111	0.023
PEP	0.224	0.059
WMT	0.003	0.0002
TSLA	0.029	0.035
NVDA	0.095	0.049
AMD	0.159	0.056
AMZN	0.192	0.063
META	0.00002	0.000001

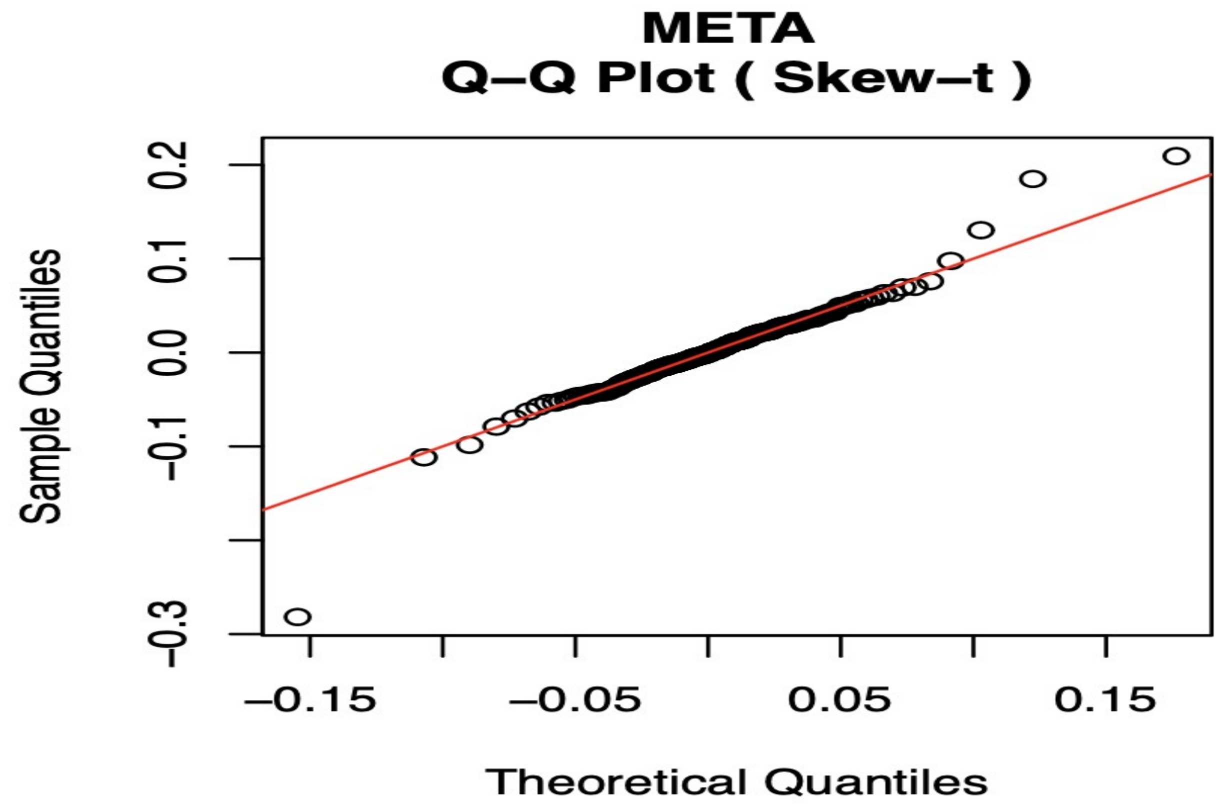
EDA (Continued) – Skewed t

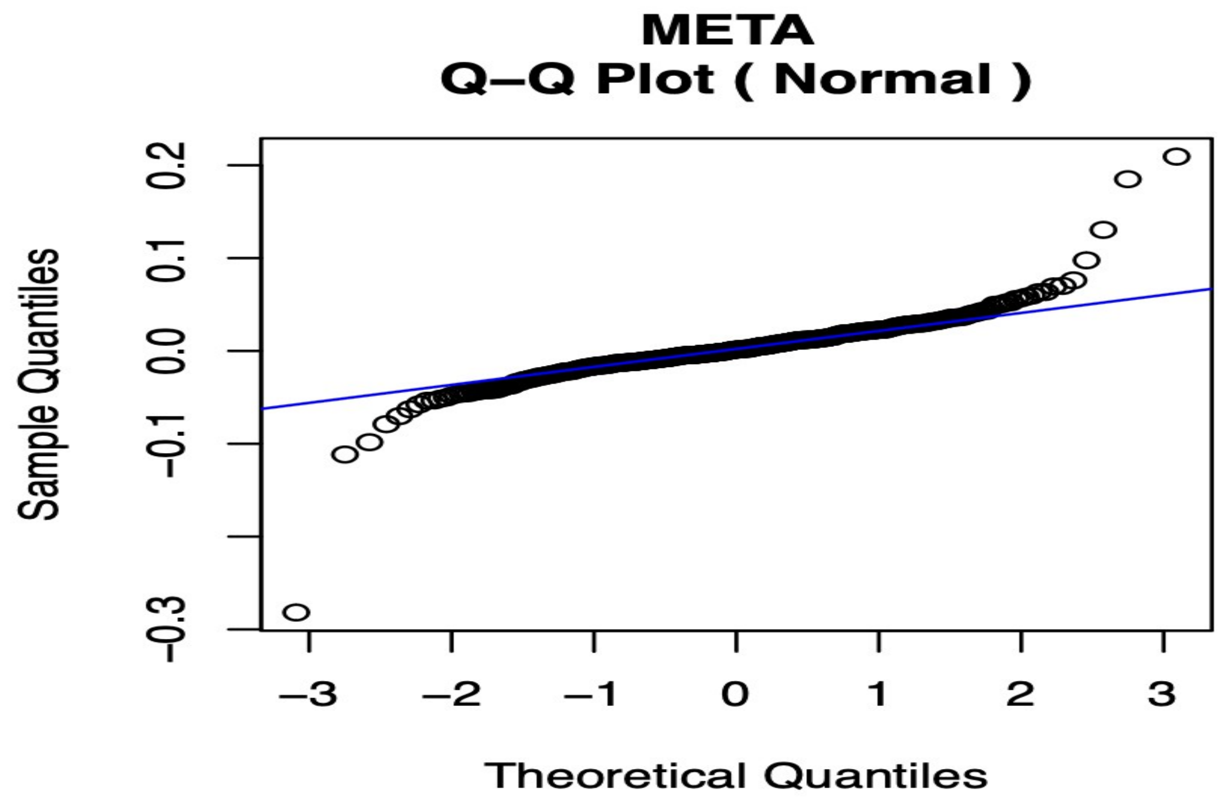
Stock (Ticker)	Kolmogorov – Smirnov Test p – value	Anderson Darling Test p – value
MSFT	0.981	0.989
JNJ	0.942	0.952
PG	0.979	0.998
PEP	0.876	0.996
WMT	0.978	0.987
TSLA	0.844	0.969
NVDA	0.997	0.997
AMD	0.846	0.989
AMZN	0.956	0.997
META	0.855	0.871

EDA (Continued) – QQ Plots









The Skewed t Hierarchical Model

- The model introduces **skewness** using a latent variable $z[i,j]$.
-
- $z[i,j] \sim Normal\ (0,2)$ and truncated at $z[i,j] > 0$, i.e. a half normal.
- The log returns themselves are modeled as:

$$y[i,j] \sim Normal\ (\mu[j] + \ \alpha[j] * z[i,j]\ , \tau[j]),$$

where,
$\mu[j]$ is the location
$\alpha[j]$ adjusts for skewness
$\tau[j]$ (precision) corresponds to scale
ν represents the degrees of freedom (heaviness of the tails)

Choice of Priors and Hyperpriors

Prior	Hyperprior	Intuition
Location Parameter: $\mu[j]$: $\mu[j] \sim \text{Normal}(\mu_0, \tau_0)$	$\mu_0 \sim \text{Normal}\ (0{,}1)$ $\tau_0 \sim Gamma\ (0.0001, 0.05)$	• This prior is neutral and allows the data to dominate unless there is strong evidence of drift.
Scale Parameter: $\tau[j]$ $\tau[j] \sim Gamma\ (a_0, b_0)$	$a_0 \sim Gamma\ (0.1, 0.5)$ $b_0 \sim Gamma\ (0.1, 0.5)$	• Hyperpriors with low values imply relatively more vague priors, allowing the data to dominate, which is suitable given the wide variability in stock volatilities. • The gamma distribution is a natural choice, and the vague hyperpriors reflect the diverse nature of financial markets.
Skewness Parameter: $\alpha[j]$ $\alpha[j] \sim \text{Normal}\ (0{,}1)$		• Financial log returns are often slightly skewed (e.g., more severe negative returns during crashes), but the extent of skewness is typically small. • A normal prior with zero mean is reasonable, as it assumes no strong bias toward skewness unless indicated by the data.
Degrees of Freedom: ν $\nu \sim Gamma\ (2\ , 0.1)$		• Financial log returns exhibit heavy tails due to extreme events like crashes. The prior reflects this by favoring small ν (heavier tails) but allows flexibility to fit lighter-tailed data. • A gamma prior with shape 2 and rate 0.1 is slightly informative, favoring heavier tails while still allowing the data to determine the degree of tail heaviness.

Prior | Hyperprior | Intuition

- This prior is neutral and allows the data to dominate unless there is strong evidence of drift.

- Hyperpriors with low values imply relatively more vague priors, allowing the data to dominate, which is suitable given the wide variability in stock volatilities.
- The gamma distribution is a natural choice, and the vague hyperpriors reflect the diverse nature of financial markets.

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Portfolio Statistics

- We consider a portfolio of 3 stocks:

a. Amazon (AMZN)
b. Nvidia (NVDA)
c. Procter & Gamble (PG)

Mean	Standard Deviation	Skewness	Kurtosis	95% Value at Risk
-0.00168	0.0003	0.219411	10.529	0.0291

Portfolio Convergence Diagnostics

- **Gelman-Rubin**

Convergence Diagnostic

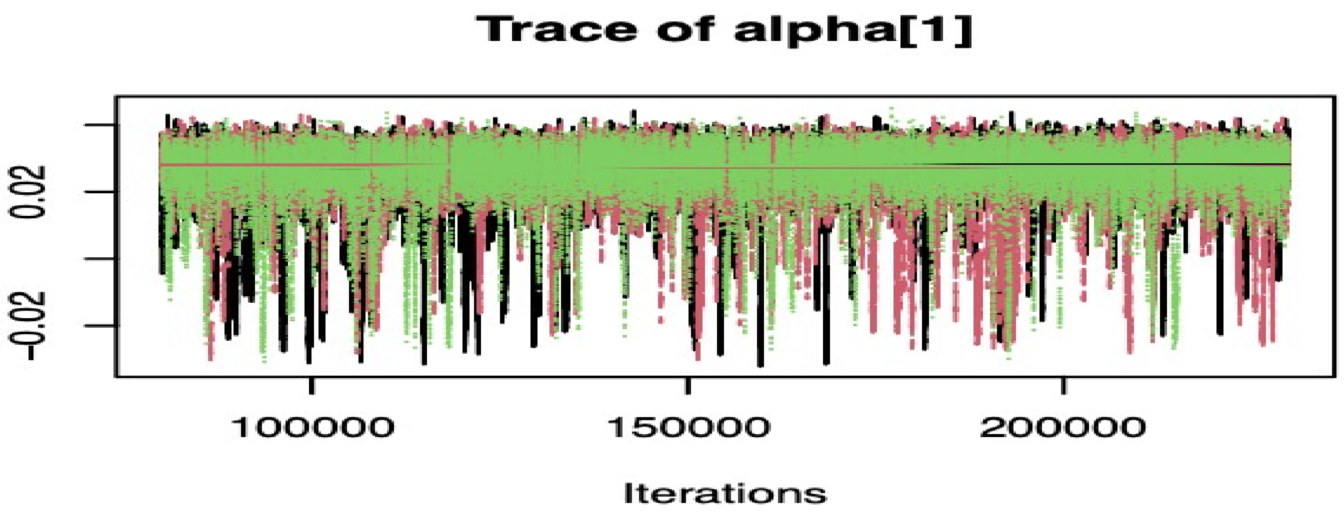
Parameter	Point Estimate	Upper C.I.
Alpha[1]	1.01	1.02
Alpha[2]	1.00	1.00
Alpha[3]	1.00	1.01
Mu[1]	1.01	1.02
Mu[2]	1.00	1.00
Mu[3]	1.00	1.01
mu0	1.00	1.00
nu	1.00	1.00
Sigma[1]	1.00	1.00
Sigma[2]	1.00	1.00
Sigma[3]	1.00	1.00
tau0	1.00	1.00

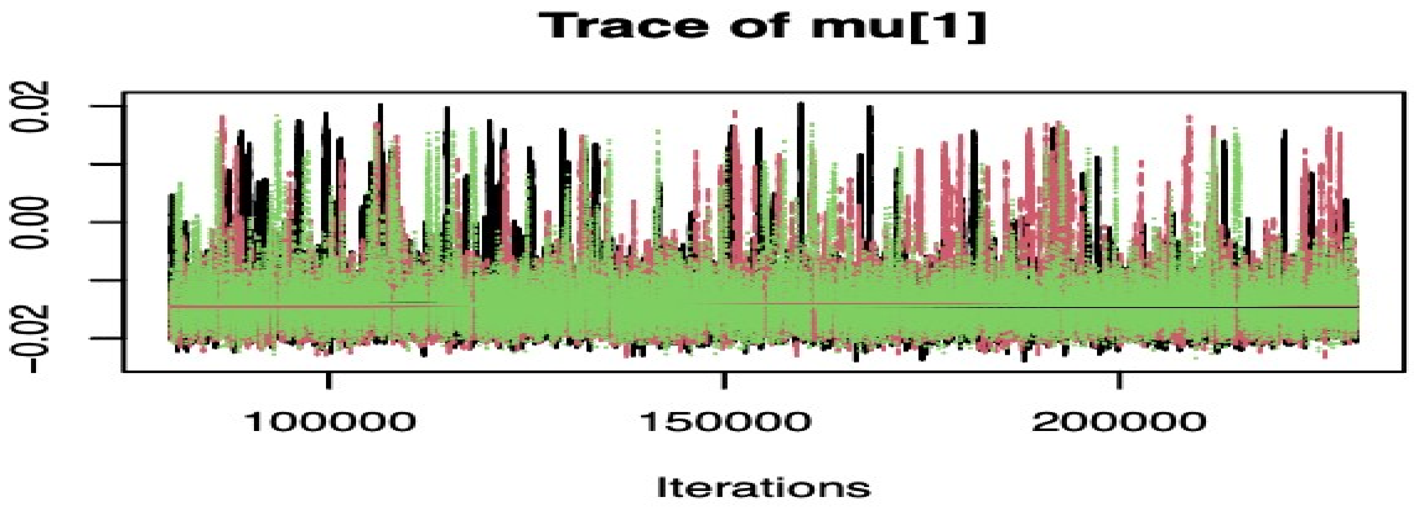
Portfolio Convergence Diagnostics

- **Effective Sample Size**

Parameter	Point Estimate
Alpha[1]	1565
Alpha[2]	15389
Alpha[3]	1699
Mu[1]	1592
Mu[2]	16475
Mu[3]	1753
Sigma[1]	4144
Sigma[2]	20940
Sigma[3]	4584

Trace Plots


Trace of alpha[1]
0.02
-0.02
100000
150000
200000
Iterations


Trace of mu[1]
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0.00
-0.02
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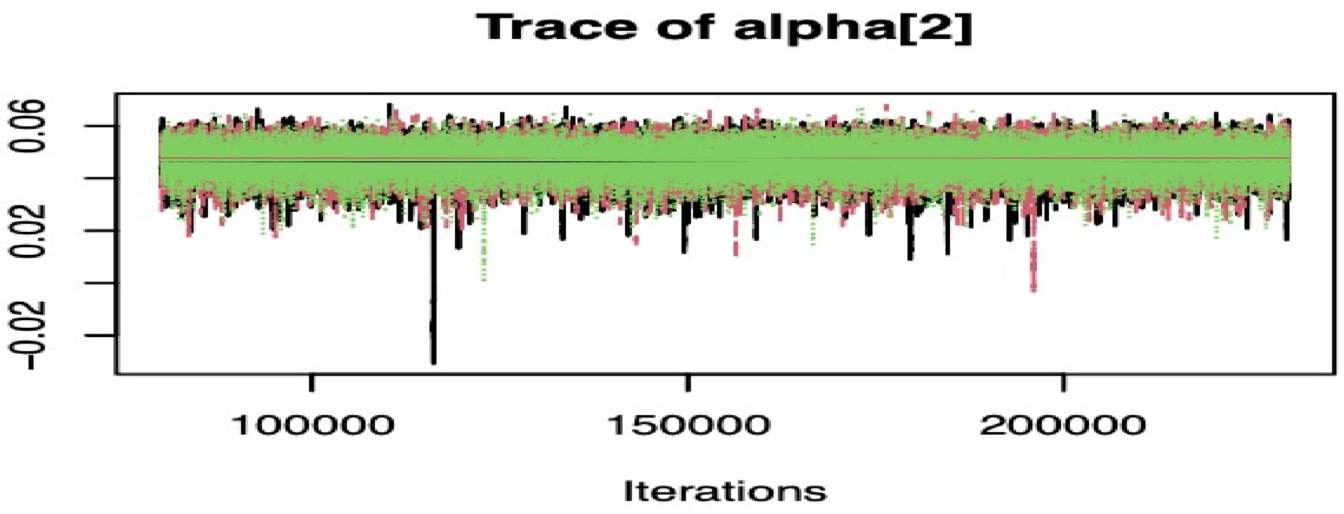
150000
200000
Iterations


Trace of alpha[2]
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-0.02
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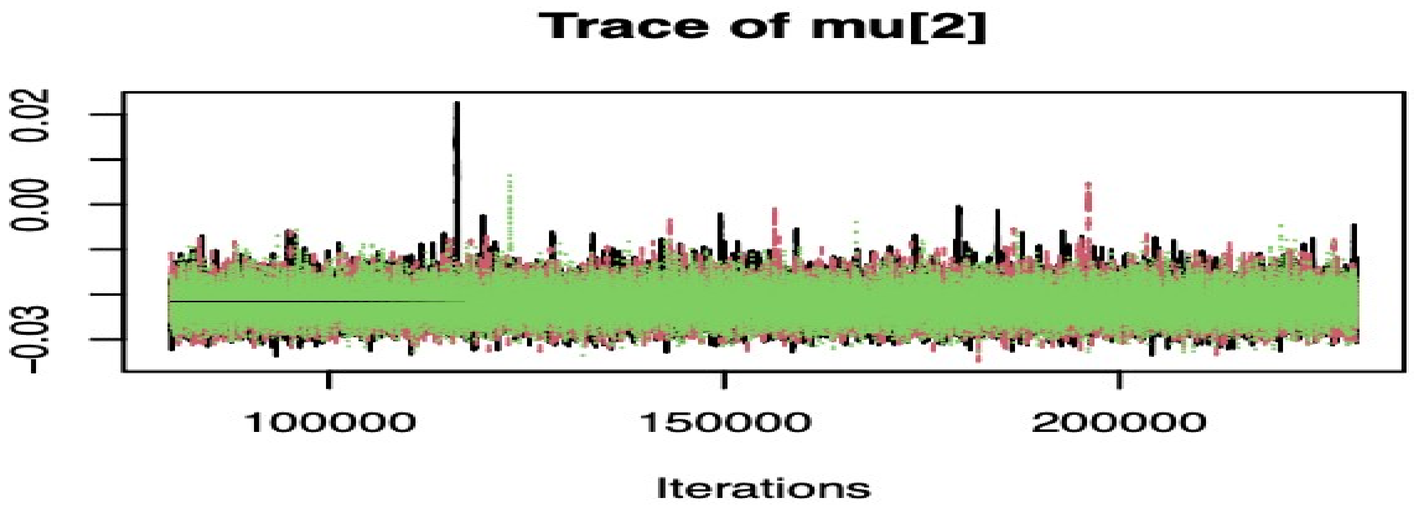
200000
Iterations


Trace of mu[2]
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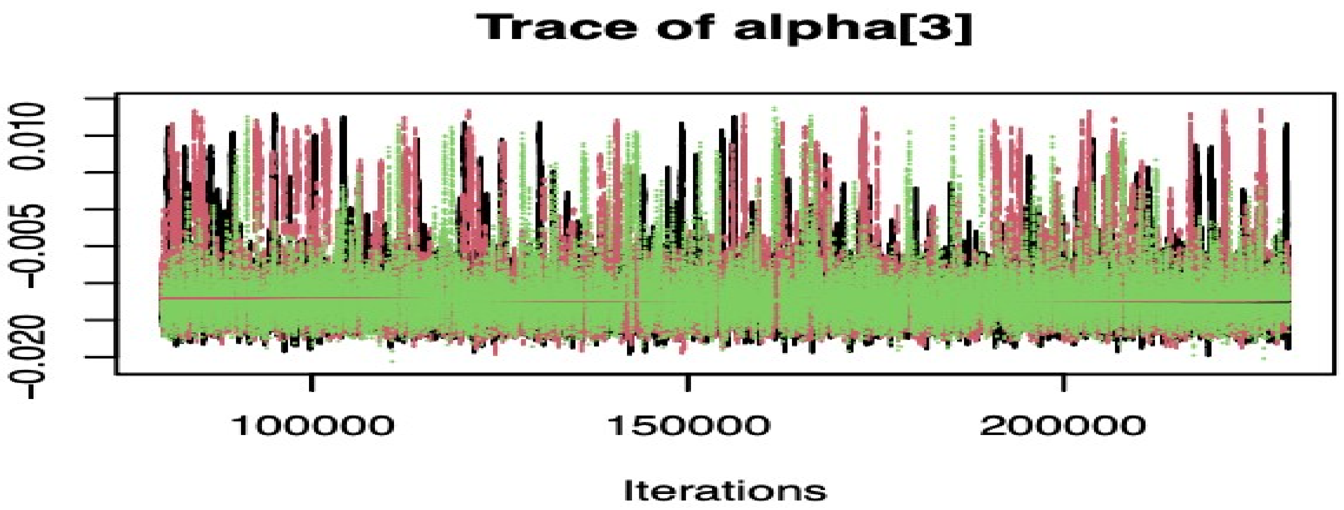
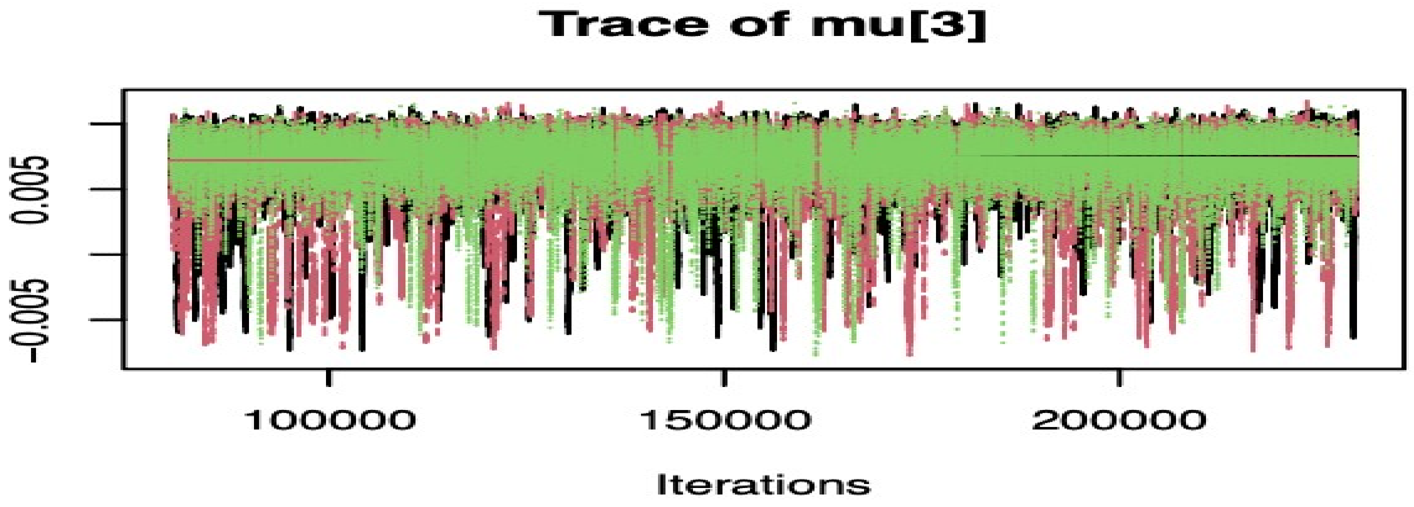
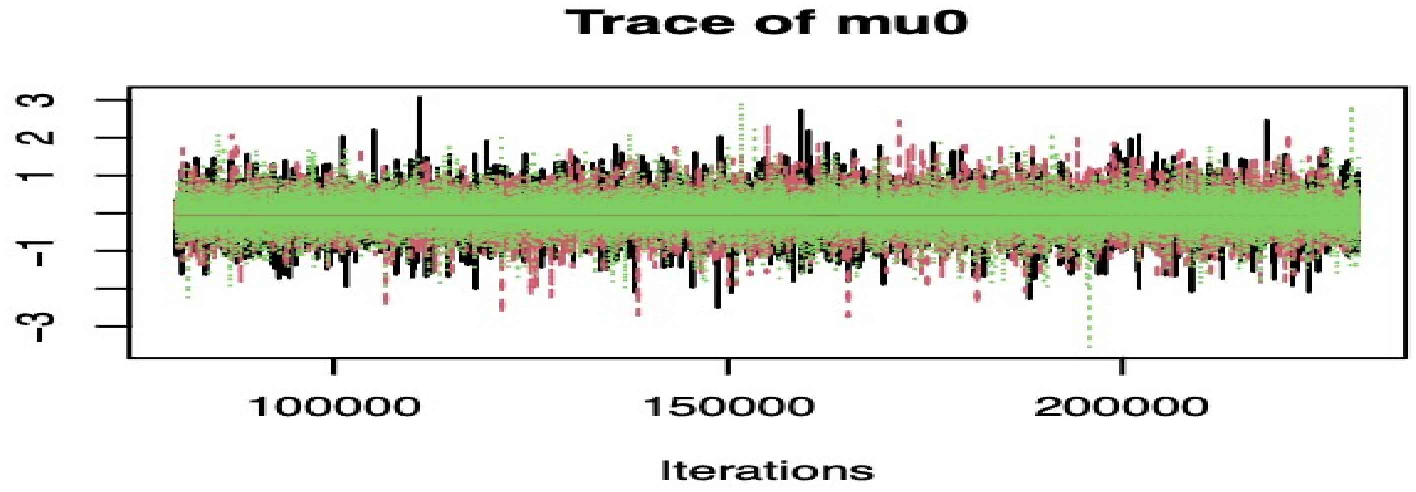
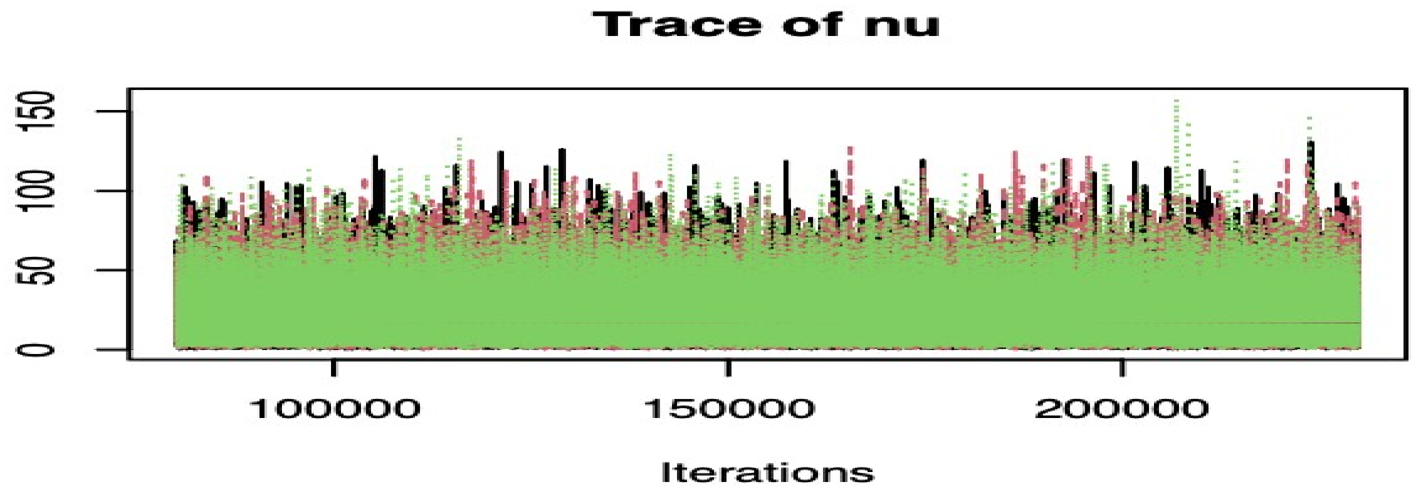
0.00
-0.03
100000
150000
200000
Iterations

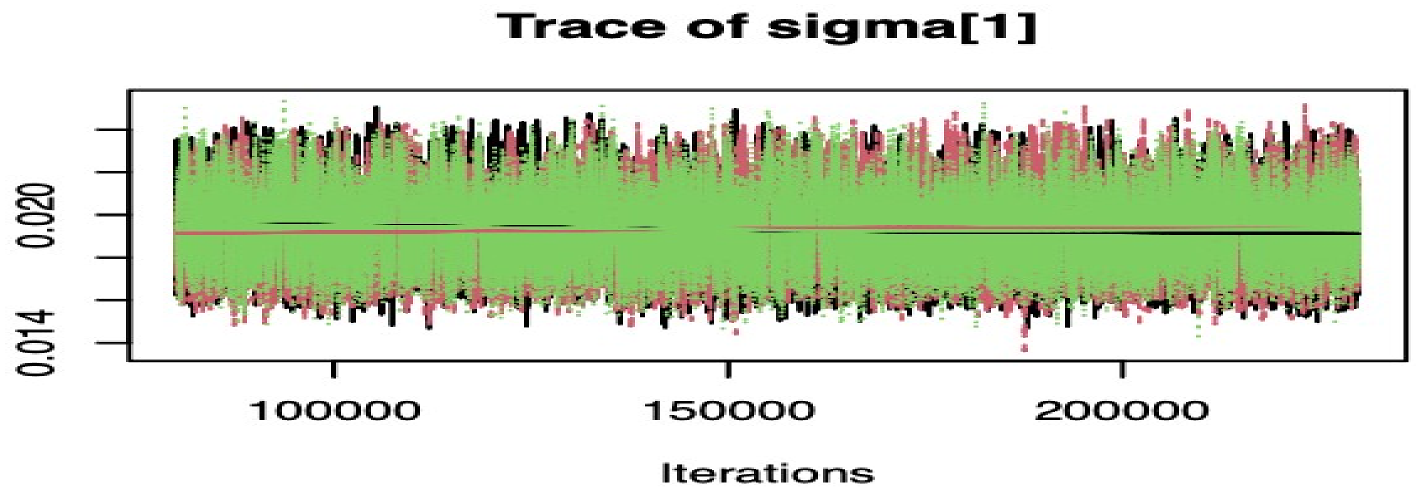

Trace of alpha[3]
0.010
-0.005
-0.020
100000
150000
200000
Iterations

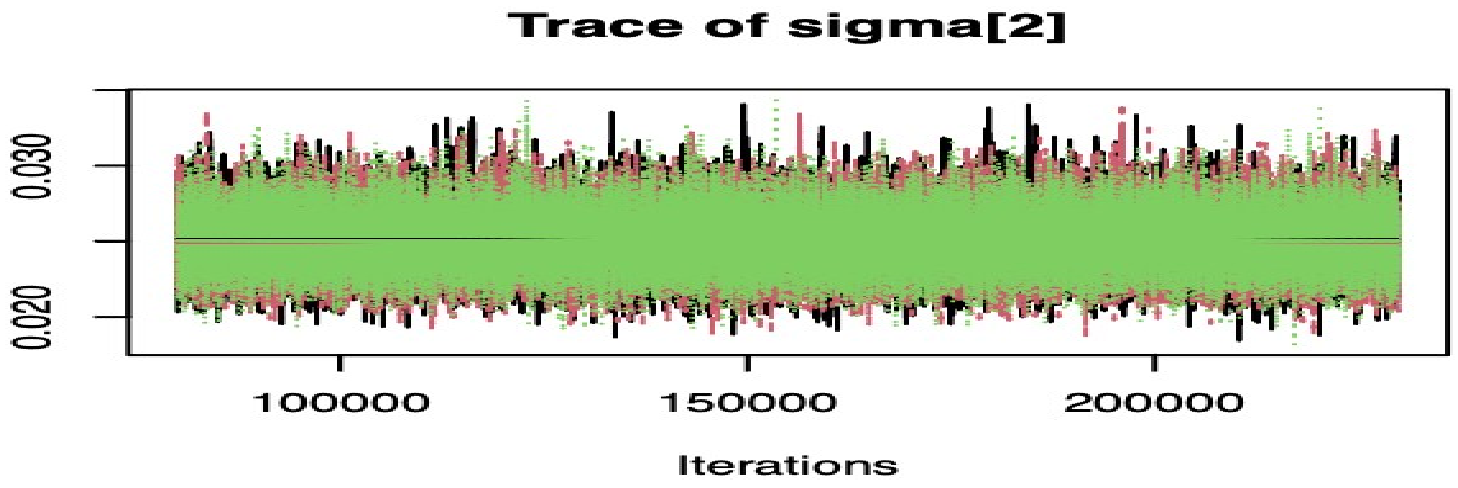

Trace of mu[3]
0.005
-0.005
100000
150000
200000
Iterations

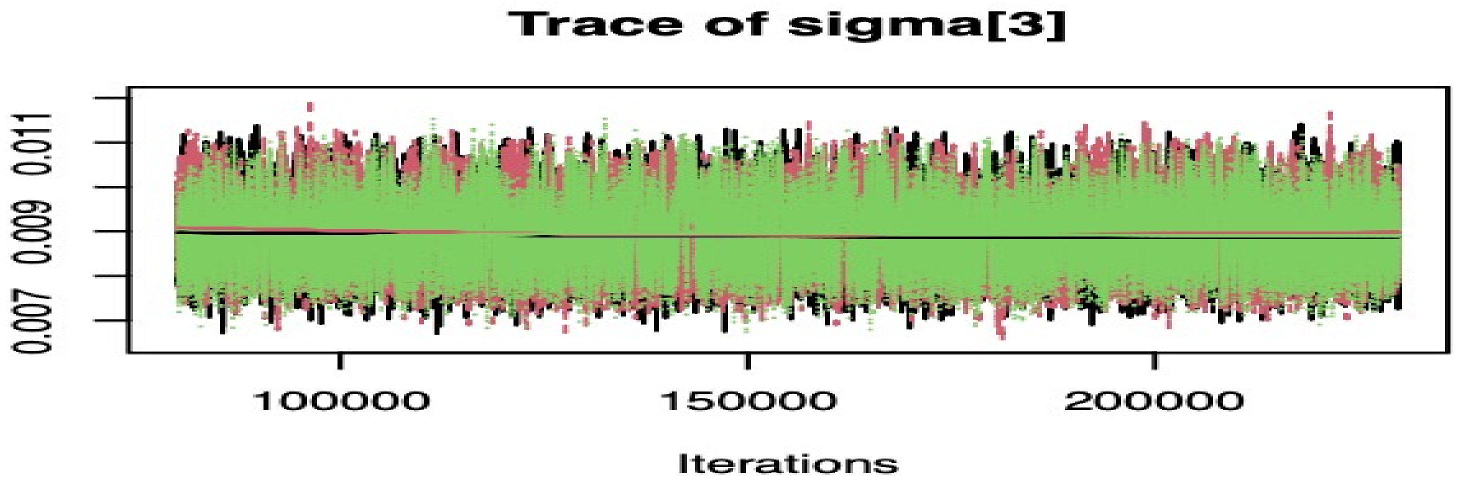
Trace Plots (Continued)

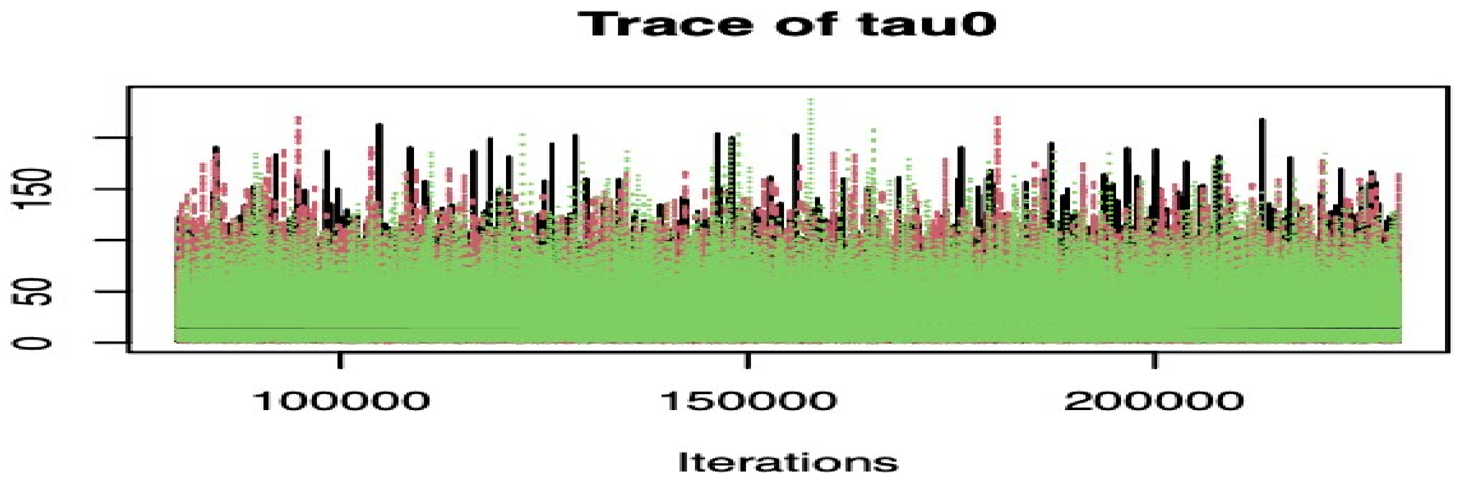












Copula Analysis

- **Copulas** are statistical tools used to model and analyze the dependence structure between random variables.
- They provide a flexible framework to model complex dependencies.
- We consider 5 copulas in our analysis:

Gaussian → t → Skewed t → Clayton → Gumbel

Copula Analysis

- Out of sample Copula Model Validation Results:

Copula	AIC	Log Likelihood
Gaussian	- 60.13	45.981
t	- 81.85	- 1573.584
Clayton	- 30.04	4.379
Gumbel	- 42.15	16.479
Skewed t	- 1444.48	69.796

Future Scope

- Use **non-centered or parametric parameterization** for hierarchical models to reduce correlation between parameters.
- Try **Hamiltonian Monte Carlo (HMC)** or **No-U-Turn Sampler (NUTS)**.
- Add **regularization priors** (e.g., Lasso or Ridge-type penalties) to shrink parameters and reduce complexity.

Thank You

Q&A